

The empirical recurrence for OEIS A276304, recovered from its tail

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Abstract

OEIS A276304 records “Empirical recurrence of order 59” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2319$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 62$. Its last coefficient is $c_{59} = -40 \neq 0$, so no further tail satisfies anything shorter; any order-59 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A276304 is “Number of $7Xn$ 0..2 arrays with no element equal to any value at offset (-2,-1) (-1,1) or (0,-1) and new values introduced in order 0..2.”. It has offset 1 and begins

365, 710, 14693, 203933, 3861469, 61239977, 1073114625, 17768699385, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Aug 28 2016 08:31:55, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2319$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2319$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 59: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 59.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 4638$ exact terms from $n = 3$ on, it returns order 59 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	3	c_{16}	275790109	c_{31}	−8708851266	c_{46}	−113534742
c_2	180	c_{17}	−1075130302	c_{32}	−668461454	c_{47}	51791438
c_3	845	c_{18}	3433811864	c_{33}	1485012638	c_{48}	−18087807
c_4	2292	c_{19}	−8782796507	c_{34}	7345193316	c_{49}	8394642
c_5	−9612	c_{20}	15535184981	c_{35}	−13059389482	c_{50}	3239769
c_6	−83198	c_{21}	−19695753832	c_{36}	6031507192	c_{51}	−3359683
c_7	−189394	c_{22}	20602433800	c_{37}	−3318564745	c_{52}	714865
c_8	−445937	c_{23}	−23840337175	c_{38}	6579907756	c_{53}	−545827
c_9	1888145	c_{24}	33676242812	c_{39}	−2700626833	c_{54}	24300
c_{10}	−1129011	c_{25}	−46143580155	c_{40}	−259469468	c_{55}	43979
c_{11}	9732930	c_{26}	58360869696	c_{41}	−1105965315	c_{56}	6969
c_{12}	−33071092	c_{27}	−70484569722	c_{42}	551088351	c_{57}	788
c_{13}	48391294	c_{28}	81933262870	c_{43}	378978490	c_{58}	−562
c_{14}	35706803	c_{29}	−72464807505	c_{44}	−156506319	c_{59}	−40
c_{15}	−140987244	c_{30}	36378636798	c_{45}	48394315		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 62$, and it is the recurrence OEIS A276304 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{59} - c_1 x^{59-1} - \dots - c_{59}$. Its constant term is $-c_{59} = 40$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 59 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 59, so $\deg q = 59$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 59, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 59, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -40 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-59 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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